

JEDEC STANDARD

Mechanical Shock – Device and Subassembly

JESD22-B110B.01

(Revision of JESD22-B110B, July 2013)

JUNE 2019

JEDEC SOLID STATE TECHNOLOGY ASSOCIATION



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TEST METHOD B110B.01 MECHANICAL SHOCK –DEVICE AND SUBASSEMBLY

(From JEDEC Board Ballot JCB-13-27, formulated under the cognizance of the JC-14.1 Subcommittee on Reliability Test Methods for Packaged Devices.)

1 Scope

Device and Subassembly Mechanical Shock Test Method is intended to evaluate devices in the free state and assembled to printed wiring boards for use in electrical equipment. The method is intended to determine the compatibility of devices and subassemblies to withstand moderately severe shocks. The use of subassemblies is a means to test devices in usage conditions as assembled to printed wiring boards. Mechanical Shock due to suddenly applied forces, or abrupt change in motion produced by handling, transportation or field operation may disturb operating characteristics, particularly if the shock pulses are repetitive. This is a destructive test intended for device qualification.

2 Apparatus

The shock-testing apparatus shall be capable of providing shock pulses with a peak acceleration of up to 2900 multiples of gravity (g), a velocity change of 100 to 544 centimeters per second (39 to 214 inches per second), and a pulse duration between 0.3 and 8.0 milliseconds to the body of the device. For free-state testing, a velocity change of 125 to 544 centimeters per second (49 to 214 inches per second), and a pulse duration between 0.3 and 2.0 milliseconds is sufficient. Conversely, for mounted-state testing, apparatus capable of a velocity change of 100 to 544 centimeters per second (39 to 214 inches per second), and a pulse duration between 5.0 and 8.0 milliseconds to the body of the device is sufficient.

The acceleration pulse shall be a half-sine waveform with an allowable deviation from specified peak acceleration not greater than $\pm 10\%$. The test velocity change shall be $\pm 10\%$ of the specified level. The pulse duration shall be measured between the points at 10% of the peak acceleration during rise time and 10% of the peak acceleration during decay time. Absolute tolerances of the pulse duration shall be $\pm 15\%$ of the specified duration. The test equipment transducer shall have a natural frequency greater than 5 times the frequency of the shock pulse being established, and measured through a low-pass filter having a bandwidth greater than 5 times the frequency of the shock pulse being established. Filtering should not be used in lieu of good measurement setup and procedure practices.

Appropriate equipment calibration should be considered prior to any testing to ensure conformance to the specified targets and acceptable tolerances. Reserving a set of known good units is recommended for pre-test calibration exercise whenever new samples are to be tested. If calibration tests are conducted regularly, then following periodical preventive maintenance should suffice for the equipment to meet the target and tolerance limits.

3 Terms and definitions

component: A constituent part.

NOTE 1 Examples include source and drain regions as components of transistors, lead frames and dice as components of packaged integrated circuits, resistors and integrated circuits as components of printed circuit boards, motherboards as components of computers, LCD screens as components of monitors, ac and dc components of complex waveforms, and loops and algorithms as components of software programs.

NOTE 2 The classification of an item as a device or a component depends upon the intention of the owner at the time of classification.

dead-bug (orientation): The orientation of a package with the terminals facing up.

device: A piece of equipment, a mechanism, or another entity designed to serve a special purpose or perform a special function.

NOTE 1 In JEDEC documents, the word “device” is often used as an abbreviated reference to the type or types of solid-state devices that are within the scope of those documents. Context could indicate otherwise; e.g., in the phrase “the device used to hold the device under test”, the first usage of the word “device” refers to a mechanism; the second to a solid-state device.

NOTE 2 The classification of an item as a device or as a component depends upon the intention of the owner at the time of classification.

deviation from specified acceleration [velocity change] [pulse duration] level: The maximum difference between the measured value and the target value.

equivalent drop height: The free-fall drop height from which an object at rest must fall, in vacuum, under standard gravity, to attain a velocity equal to the velocity change stated in the test specification.

NOTE This is the theoretical height that will impart the specified velocity change if impact with zero rebound occurs. This height is provided for reference only in the various service conditions.

free state: The state in which a device or subassembly is rigidly attached to the test apparatus so that the full specified shock level is transmitted to the device or subassembly body.

live-bug (orientation): The orientation of a package with the terminals facing down, e.g., resting on its terminals.

mounted state: The state in which a subassembly is supported by a test fixture that allows flexure to simulate usage conditions and in a manner such that the full specified shock level is transmitted to the subassembly body.

peak acceleration: The maximum acceleration during the dynamic motion of the sample under test.

pulse duration: The time interval between the instant when the acceleration first reaches 10% of its specified peak level and the instant when the acceleration first returns to 10% of the specified peak level after having reached that peak level.

NOTE The basic frequency of the pulse is $1/(2 \times \text{duration})$.

3 Terms and definitions (cont'd)

service condition: The designation for the severity of stress.

subassembly: A printed circuit board and the devices assembled thereon that form a unit or segment of electrical equipment.

NOTE The devices are preferably located near the center of the printed circuit board.

velocity change: The integral of the acceleration interval over the duration of the entire shock event including at least the pulse duration interval.

vertical direction: The direction that is parallel with gravity, i.e., normal to the normalized surface of the earth.

Diamond Foundry

4 Procedure

The shock-testing apparatus shall be attached to a sturdy laboratory table or equivalent base and leveled before use. Means shall be provided to prevent the shock from being repeated due to “bounce” in the apparatus. In the free state unless otherwise specified, the device or subassembly shall be subjected to a total of 30 shocks, which are five shock pulses of the peak acceleration, velocity change and pulse duration specified in the selected service condition (see Table 1) in each of the positive and negative directions of three orthogonal axes (X, Y and Z). If shock testing is required in the mounted state, the subassembly shall be subjected to a total of 12 shocks, which are two shock pulses of the peak acceleration, velocity change and pulse duration specified in the selected service condition (see Table 2) in each of the positive and negative directions of three orthogonal axes (X, Y and Z). Figure 1 and Figure 2 define the device orientation, positive and negative directions for the three orthogonal axes.

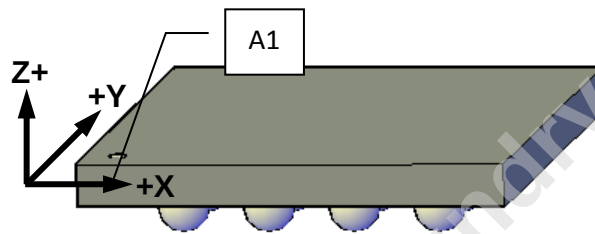


Figure 1 — Live bug orientation with solder spheres of device facing downward in either free or mounted state.

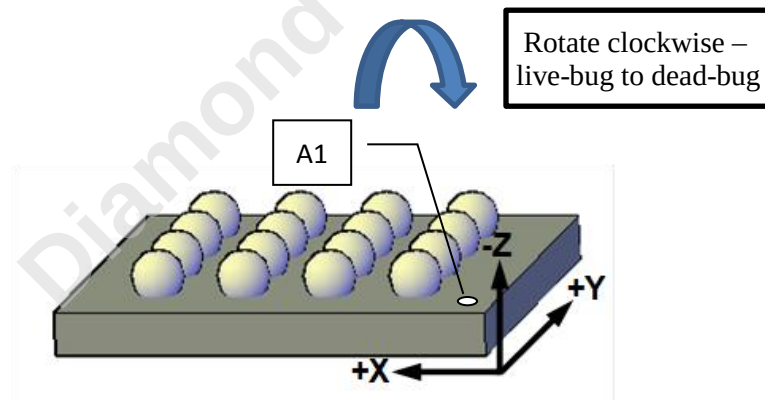


Figure 2 — Dead bug orientation with solder spheres of device facing upward in either free or mounted state.

There are two types of tests that can be performed, based on the test information needed. One is testing of a device or subassembly in the free-state, and another is testing of a subassembly in a mounted-state.

Devices or subassemblies subjected to the test shall be randomly selected and typical of production. In the free-state, the device or subassembly shall be rigidly attached to the test apparatus, so that the full specified shock level is transmitted to the device or subassembly body. If the mounted state test is performed, the method of mounting to the test apparatus shall be typical of the usage condition. If device or subassembly rework, reuse, remounting, burn in, or other stressful process is possible, such a process or processes should be applied to the device or subassembly prior to the shock test. Use of such processes in the test hardware preparation shall be documented.

Test Method B110B.1

(Revision of Test Method B110B)

4.1 Device or subassembly in free-state

Devices or subassemblies to be tested in the free- state must be subjected to at least one of the service conditions (A to H) shown in Table 1, which shall be documented. The designated shock shall be applied to the device or subassembly body in a manner to simulate expected impacts during processing, packaging, and shipment. The device or subassembly shall be rigidly attached to the test apparatus in such a manner that it experiences the full-specified shock level at the device or subassembly body. At least five shocks in each of two directions of three orthogonal axes shall be applied (minimum total of 30 shocks) at the severity of the designated service condition.

Table 1 — Device or subassembly free state test levels

Service condition	Acceleration peak	Pulse duration	Velocity change		Equivalent drop height	
	g	ms	cm/s	in/s	cm	inches
H	2900	0.3	543	214	150	59
G	2000	0.4	499	197	127	50
B	1500	0.5	468	184	112	44
F	900	0.7	393	155	78.9	31
A	500	1.0	312	123	49.7	20
E	340	1.2	255	100	33.1	13
D	200	1.5	187	73.7	17.9	7
C	100	2.0	125	49.2	7.90	3
NOTE velocity change = $2 * (\text{std_g} * \text{acceleration_peak_g} * \text{pulse_duration}) / \pi$ drop height = $0.5 * (\text{velocity_change})^2 / \text{std_g}$.						

4.2 Subassembly in mounted state

If required, subassemblies shall also be tested in a mounted state, subject to at least one of the service conditions (1 to 14) shown in Table 2, which shall be documented. The designated shock shall be applied to the subassembly mounted to the test apparatus with fixtures that allow flexure to simulate the usage conditions, and in a manner such that the full specified shock level is transmitted to the subassembly body. Preferred methods to support the subassembly are a slotted/clamping 'picture frame', or a raised-boss bolted fixture with contact points in four regions not closer than one inch from any device. The subassembly, supporting method, test fixture mounting dimensions, and one or more of the lowest resonant frequencies of the subassembly shall be documented. At least two shocks in each of two directions of three orthogonal axes shall be applied (minimum total of 12 shocks) at the severity of the designated service condition.

The optimum test is performed when the subassembly is mounted in a manner that simulates the application configuration. If that information is unknown or unavailable, a recommended printed wiring board for testing the device(s) of the subassembly is the JEDEC standard thermal card (as described in JEDEC Standards JESD51-9, JESD51-10, and JESD51-11), which should be modified to include device and connection functionality circuitry.

4.2 Subassembly in mounted state (cont'd)

Alternately, if the size or construction of the JEDEC thermal card is not suitable for the given device(s), a printed wiring board should be used with the dimensions, materials, and construction typical for the device subassembly usage, and should include electrical circuitry to test for functionality, continuity, and damage of the device on the subassembly. The dimensions and construction of the printed wiring board shall be documented. Test results using subassembly application hardware are most relevant and should take precedence over results obtained using test vehicles.

Table 2 — Subassembly mounted state test levels

Service condition	Acceleration peak	Pulse duration	Velocity change		Equivalent drop height	
	g	ms	cm/s	in/s	cm	inches
1	109	8.0	544	214	151	60
2	108	7.5	506	199	130	51
3	107	7.0	468	184	111	44
4	105	6.5	426	168	92.6	36
5	103	6.0	386	152	75.9	30
6	95	5.8	344	135	60.3	24
7	86	5.6	301	118	46.1	18
8	72	5.4	243	95.6	30.0	12
9	67	5.3	222	87.3	25.1	10
10	61	5.2	198	78.0	20.0	8
11	54	5.1	172	67.7	15.1	6
12	45	5.0	140	55.3	10.1	4
13	39	5.0	122	47.9	7.6	3
14	32	5.0	100	39.3	5.1	2
NOTE velocity change = $2 * (\text{std_g} * \text{acceleration_peak_g} * \text{pulse_duration}) / \pi$ drop height = $0.5 * (\text{velocity_change})^2 / \text{std_g}$.						

4.3 Measurements

Hermeticity tests, if required, visual examination and electrical measurements (consisting of parametric and functional tests) shall be performed.

5 Failure criteria

A Device or subassembly shall be considered as a failure if hermeticity requirements, if any, cannot be demonstrated, if parametric limits are exceeded or if functionality cannot be demonstrated under the conditions specified in the applicable procurement document. Mechanical damage, such as cracking, chipping or breaking of portions of the device shall also be considered a failure provided such damage was not caused by fixturing or handling and the damage is critical to performance in the specific application. Subassembly failure criteria may include, but is not limited to, solder connections to the printed wiring board, compliant pin leads, adhesive, encapsulation, or underfill that materially affect device or subassembly reliability.

6 Summary

The following details shall be specified in the applicable procurement document:

- a) Test service condition, for each test performed.
- b) Electrical measurements.
- c) Sample size and accept number.
- d) Disposition of failures.
- e) Hermetic leak rate (if applicable).
- f) Description of mounted state test vehicle and fixture (if applicable).
- g) Description of device pre-test stress history (if applicable).

Annex A (informative) Differences between JESD22-B110B and JESD22-B110A

A.1 Differences between JESD22-B110B.01 and JESD22-B110B

The following list briefly describes the changes made to entries that appear in this publication, JESD22-B110B.01, compared to its predecessor, JESD22-B110B (TBD, last reaffirmed June 2009).

Clause	Description of Change
All	Replacement of Component with Device in most areas to be consistent with JEDEC JESD88
3	Add Device in Terms and Definitions
3	Clarified free state setup as appropriate for device and/or subassembly

A.2 Differences between JESD22-B110B and JESD22-B110A

The following list briefly describes most of the changes made to entries that appear in this publication, JESD22-B110B, compared to its predecessor, JESD22-B110A (November 2004, reaffirmed June 2009).

Clause	Description of Change
All	Combined Mechanical Shock JESD22-B104C into this standard to establish one document covering unattached components and components attached to printed wiring boards.
2	Modified tolerance limits; peak acceleration reduced from $\pm 20\%$ to $\pm 10\%$; pulse duration reduced from $\pm 30\%$ to $\pm 15\%$.
3	Updated Terms and definitions.
4	Added Figures 1 and 2.
4.1	Updated Table 1: mathematical formulas added for velocity change and equivalent drop height; calculated values shown in table.
4.2	Updated Table 2: mathematical formulas added for velocity change and equivalent drop height; calculated values shown in table; portable subassembly conditions removed.



Standard Improvement Form**JEDEC JESD22-B110B.01**

The purpose of this form is to provide the Technical Committees of JEDEC with input from the industry regarding usage of the subject standard. Individuals or companies are invited to submit comments to JEDEC. All comments will be collected and dispersed to the appropriate committee(s).

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1. I recommend changes to the following:

☐ Requirement, clause number _____

☐ Test method number _____ Clause number _____

The referenced clause number has proven to be:

☐ Unclear ☐ Too Rigid ☐ In Error

☐ Other _____

2. Recommendations for correction:

3. Other suggestions for document improvement:

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